

SLC7A5 (human + orthologs) · 1 responding molecules · 4 responding interaction partners · 0 upstream kinases · showing 26 of 31 edges (max 5 per node) · 20 nodes · 26 edges

Rings layout

Fit

Hypotheses 3

Details

Ask Claude

Generated from what the graph shows right now. Green cards summarise evidence already in the graph; amber cards are new hypotheses built by combining it.

From the graph

SLC7A5: human up, no rat change

Under current filters (species rat + human; organs Rat Gastrocnemius, Rat Vastus lateralis, Human Muscle (VL)), [D1, D2, D3, D4, D8, M2, M1]

- Human: 3 responses in Human Muscle (VL); layers Transcript; max [z] 6.7 [D3, D4, D8, M2, M1]
- Rat: none [D1, D2]

► References (7)

New hypothesis

SLC7A5 sits in a responding interaction module

4 physical interaction partners also change (3 mostly up, 1 mostly down). [R1, R2, D1, D2, D3, D4, D8, M2, M1] Co-regulated interaction partners suggest a complex or signalling module that exercise remodels as a unit. [L7, R3]

- UBC: human down, rat up [R1, R2, D1, D2, D3, D4, D8, M2, M1]
- SLC3A2: human up, rat – [R1, R2, D1, D2, D3, D4, D8, M2, M1]
- SLC7A8: human up, rat – [R1, R2, D1, D2, D3, D4, D8, M2, M1]
- NOS2: human up, rat – [R1, R2, D1, D2, D3, D4, D8, M2, M1]

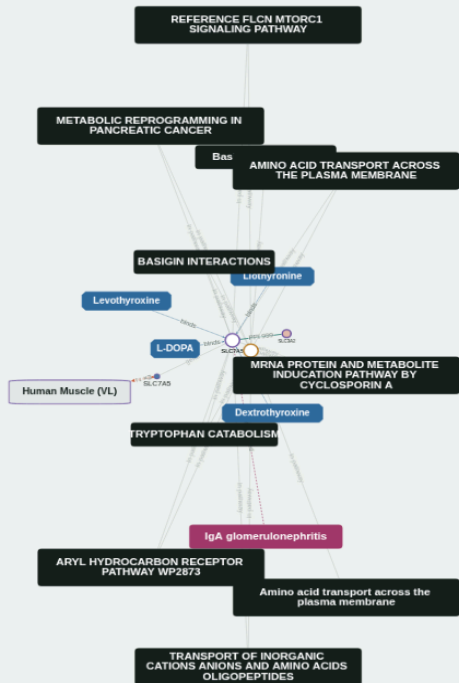
Test: Co-immunoprecipitate SLC7A5 with UBC and SLC3A2 before and after exercise (or contraction in vitro) to see whether the complex changes. [R3]

► References (11)

UBC SLC3A2 SLC7A8 NOS2

From the graph

- Human
- Muscle Adipose Blood All
- Vastus lateralis
- Heart
- Kidney
- Blood (RNA)
- Cortex
- Hypothalamus
- Colon
- Spleen
- Testes
- Vena cava
- Adipose (SC)
- All
- 2 wk 4 wk 8 wk
- during 40 min
- post 15–45 min
- post 24 h



Exercise KG Explorer

MoTrPAC rat endurance training · human acute exercise · significant responses · protein interactions

Pathway

Gene

Metabolite

Phenotype

Disease

Theme

Choose...



DISEASE

IgA glomerulonephritis

SPECIES

Rat Human

ORGAN

Muscle Adipose Blood All

Gastrocnemius Vastus lateralis

Subcut. white fat Heart

Liver Lung Kidney

Brown fat Blood (RNA)

Plasma Cortex

Hippocampus Hypothalamus

Adrenal Colon

Small intestine Spleen

Ovary Testes Vena cava

Muscle (VL) Adipose (SC)

Blood

TIMEPOINT

All

1 wk 2 wk 4 wk 8 wk

during 20 min during 40 min

post 10 min post 15–45 min

post 3.5–4 h post 24 h

SEX (RAT)

Female Male

DIRECTION

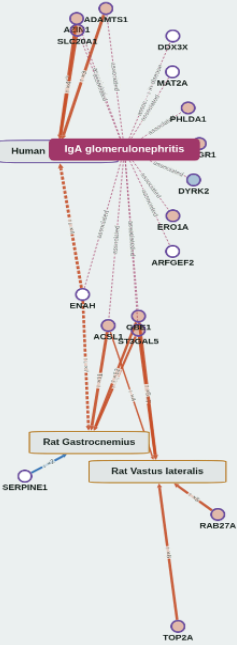
Up Down

OMICS LAYER

IgA glomerulonephritis (Hetionet) · 204 associated genes respond · genes over-represented in 0 organ(s) · direction test significant in 1 · 0 drugs bind shown genes · showing 29 of 93 edges (max 5 per node) · 21 nodes · 29 edges

Rings layout

Fit



Hypotheses 1

Details

Ask Claude

Generated from what the graph shows right now. Green cards summarise evidence already in the graph; amber cards are new hypotheses built by combining it.

New hypothesis

Exercise pushes IgA glomerulonephritis genes against the disease direction

In Human Muscle (VL), most genes that are dysregulated in IgA glomerulonephritis move the opposite way with exercise. [D3, D4, D8, M2, M1, R2, R19, P1] Reversing a disease expression signature is one route by which exercise could act as therapy. [L16]

- Human Muscle (VL): 185 of 267 genes opposite (69%), adj p 3.8e-8 [R2, R19, M1, P1]

Test: Check whether these opposing genes also move toward normal in patient tissue after a training program, and whether the shift scales with fitness gains. [L16, P1]

► References (9)

CLAUDE

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Exercise KG Explorer

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Pathway

Gene

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Disease

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Choose...

GENE

SLC7A5

SPECIES

Rat

Human

ORGAN

Muscle

Adipose

Blood

All

Gastrocnemius

Vastus lateralis

Subcut. white fat

Heart

Liver

Lung

Kidney

Brown fat

Blood (RNA)

Plasma

Cortex

Hippocampus

Hypothalamus

Adrenal

Colon

Small intestine

Spleen

Ovary

Testes

Vena cava

Muscle (VL)

Adipose (SC)

Blood

TIMEPOINT

All

1 wk

2 wk

4 wk

8 wk

during 20 min

during 40 min

post 10 min

post 15–45 min

post 3.5–4 h

post 24 h

SEX (RAT)

Female

Male

DIRECTION

Up

Down

OMICS LAYER

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Fit

Amino acid transport across the plasma membrane

TRANSPORT OF INORGANIC CATIONS ANIONS AND AMINO ACIDS OLIGOPEPTIDES

TRYPTOPHAN CATABOLISM

AMINO ACID TRANSPORT ACRO THE PLASMA MEMBRANE

ARYL HYDROCARBON RECEPTOR PATHWAY WP2673

Levothyroxine

L-DOPA

Basigin interactions

Liothyronine

METABOLIC REPROGRAMMING IN PANCREATIC CANCER

IgA glomerulonephritis

Human Muscle (VL)

REFERENCE FLCN MTORC1 SIGNALING PATHWAY

ETABOLITE MAY BY JA

Hypotheses 3

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SLC7A8: human up, rat – [R1, R2, D1, D2, D3, D4, D8, M2, M1]

NOS2: human up, rat – [R1, R2, D1, D2, D3, D4, D8, M2, M1]

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UBC

SLC3A2

SLC7A8

NOS2

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Human Muscle (VL)

IgA glomerulonephritis

Rat Vastus lateralis

Rat Gastrocnemius

SERPINE1

PHLDA1

SLC3A1

ADAMTS1

DDX3X

ARFGEF2

MAT2A

IFNGR1

DTRK

ERO1A

AZIN1

ENAH

GAB2

GAT5

ACS11

RAB27A

TOP2A

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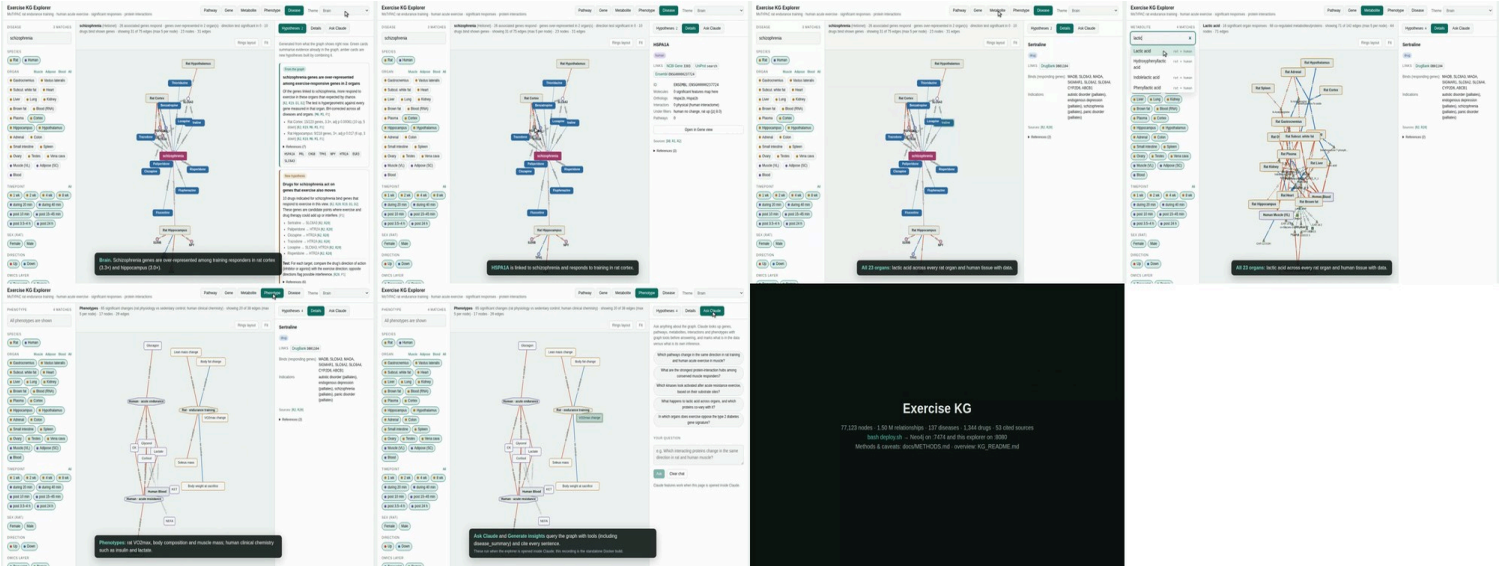
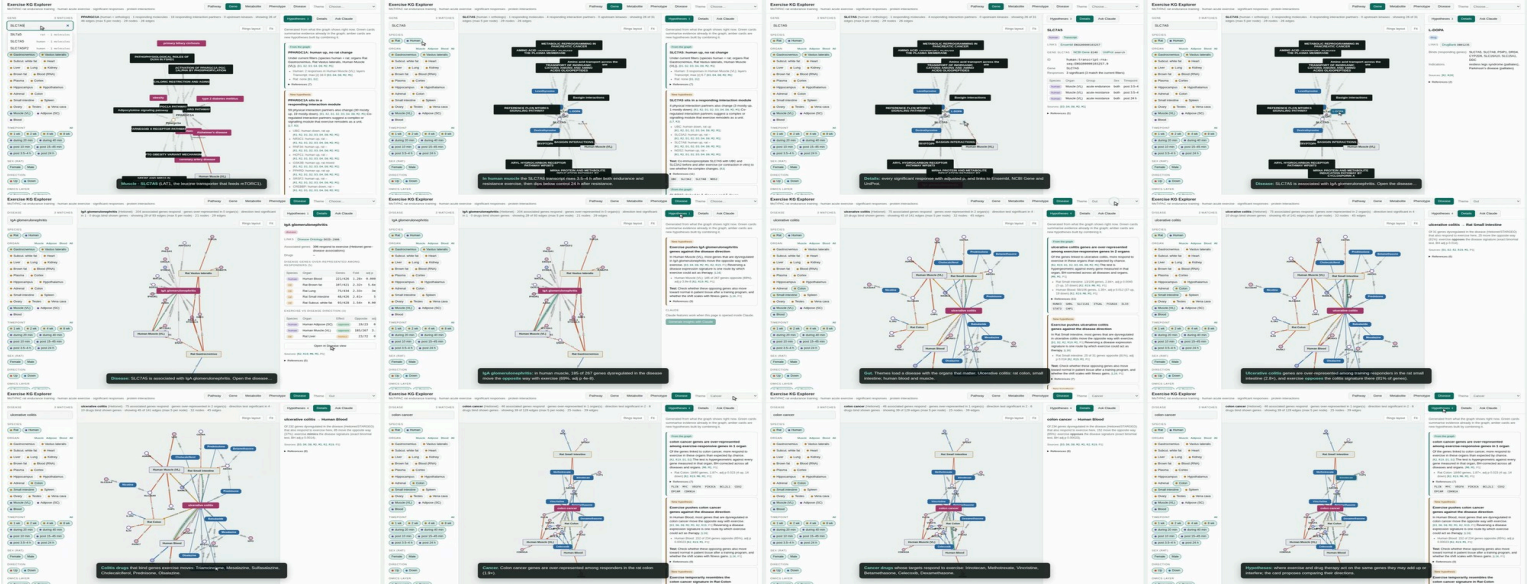
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Exercise KG

77,123 nodes - 1,561 bi-relationships - 137 datasets - 1,541 drugs - 14 cited sources
Search display on - Termset 1014 and this explorer on 3002
Network 5 sources: BioGRID, ChEMBL, DrugBank, KEGG, UniProt